

CUT Embedded Wall

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1. General

CUT Embedded Wall is a specialized analytical–computational framework centered on advanced analytical earth-pressure formulations and their coupling with elastic beam-based soil–structure interaction analysis for embedded retaining walls.

Its philosophy is to combine:

- the transparency, robustness, computational efficiency, and reliability of analytical methods,
- with the realism and engineering confidence commonly associated with finite-element analysis.

In this way, CUT Embedded Wall seeks to provide FEM-level engineering insight without forcing the user into a heavy black-box numerical environment.

2. Highlights

- Analytical–computational platform for embedded wall analysis
- Powered by the Pantelidis Generalized Coefficients of Earth Pressure framework
- Unified treatment of at-rest, active, passive, and intermediate mobilized earth pressures
- Displacement-coupled mobilized earth-pressure formulations
- Cohesive–frictional soil modeling
- Static and pseudo-static seismic earth-pressure analysis
- Advanced nonlinear variational beam finite-element solver

3. Wall Analysis Capabilities

3.1. Solver capability matrix

Comparative overview of the structural formulations, interaction capabilities, and optimization philosophies of the available solver models.

Feature / Capability	Closed-form bending	Differential equation	Rigid wall	General case	General variational
Wall bending (EI)	✓	✓	—	✓	✓
Rigid translation	—	—	✓	✓	✓
Rigid rotation	—	—	✓	✓	✓
Differential-equation beam solve	—	✓	—	Partial	✓
Finite-element beam formulation	—	—	—	—	✓
Rotational DOFs	—	Limited	—	Partial	✓
Nonlinear CUT pressure updating	✓	✓	✓	✓	✓
Displacement-coupled soil response	✓	✓	✓	✓	✓
Reinforcement/support interaction	Basic	✓	—	Limited	✓
Nonlinear support activation	—	Partial	—	Partial	✓
Plastic support capping	—	—	—	Partial	✓

Feature / Capability	Closed-form bending	Differential equation	Rigid wall	General case	General variational
Energy/work admissibility	—	—	—	✓	✓
Base restraint flexibility	Fixed	Fixed	Free/rigid	Variable	Variable
Staged interaction capability	Limited	Moderate	Limited	High	High
Computational cost	Very low	Moderate	Very low	High	High
Numerical robustness	High	High	High	Moderate	High
Recommended use	Rapid checks	Engineering analysis	Preliminary assessment	Advanced interaction studies	Full nonlinear SSI analysis

3.2. Rigid wall (No bending)

Rigid-body embedded wall model neglecting wall bending deformation while retaining displacement-dependent mobilized earth-pressure interaction through wall translation and rotation.

This is the pure rigid-body solver. The wall has translation, rotation, and pivot depth, but no EI bending deformation.

Displacement field:

$$w(z) = dx + \tan(\theta)(z_{pivot} - z)$$

The solver recalculates CUT mobilized pressures from the current rigid movement and searches for equilibrium.

Optimization:

- search in dx
- search in θ
- search in z_{pivot}

Optimization hierarchy:

1. force equilibrium
2. moment equilibrium
3. minimum work band
4. minimum rotation
5. minimum translation

This is essentially a kinematic equilibrium solver rather than a beam solver.

3.3. Fixed base — closed-form bending

Analytical elastic beam solution using closed-form bending formulations coupled with displacement-dependent mobilized earth-pressure distributions.

This is the classical flexible cantilever wall model.

The wall base is fully fixed:

$$\begin{aligned}w(H_R) &= 0 \\ \theta(H_R) &= 0\end{aligned}$$

The beam deflection is obtained from analytical beam influence functions using Gauss integration.

Iterative procedure:

1. Assume wall deflection $w(z)$
2. Compute mobilized CUT pressures
3. Solve beam bending
4. Update wall deflection
5. Repeat until convergence

Reinforcement:

Anchors and props are introduced as nodal support reactions rather than pressure spikes.

Optimization:

This solver is a nonlinear fixed-point iterative solver with under-relaxation.

Convergence criterion:

$$\max |w_{n+1} - w_n| < tolerance$$

3.4. Fixed base — differential equation

Advanced elastic beam model based on differential-equation soil–structure interaction analysis with displacement-coupled mobilized earth-pressure updating and nonlinear reinforcement/support interaction capabilities.

This is the differential-equation version of the fixed-base beam.

It includes:

- flexible beam behavior
- fixed base

- nonlinear CUT pressures
- reinforcement compatibility

At every iteration:

1. Current wall displacement produces mobilized pressures
2. The beam differential equation is solved
3. Wall displacements are updated
4. The process repeats until convergence

Optimization:

Nonlinear equilibrium iteration with monitoring of:

- displacement changes
- pressure changes
- support force changes

3.5. General case

Unified embedded wall framework combining bending deformation with rigid-body translation and rotation for complex staged wall–soil interaction problems involving displacement-coupled mobilized earth-pressure distributions.

This is the hybrid variational/mechanism framework.

It combines:

- flexible bending
- rigid-body movement
- work/energy admissibility

Step 1 — Generate bending schemes:

Multiple fixed-base bending states are generated by scaling all loads from 0% to 100%.

Total displacement:

$$w_{total} = w_{bending} + w_{rigid}$$

Step 2 — Rigid search after bending:

For every bending scheme, the solver searches:

- dx
- θ
- z_{pivot}

Each candidate:

1. Builds the total wall displacement
2. Recalculates CUT mobilized pressures

3. Evaluates force equilibrium
4. Evaluates moment equilibrium
5. Evaluates work and energy compatibility

Energy admissibility:

$$W_{bend} \approx U_{bend}$$

with:

$$U_{bend} = \frac{1}{2} \int EI \kappa^2 dz$$

Optimization hierarchy:

1. Force equilibrium
2. Moment equilibrium
3. Energy admissibility
4. Least two-sided work
5. Smallest mechanism intensity

3.6. General variational

Advanced nonlinear beam finite-element framework combining Euler–Bernoulli beam bending, rigid-body translation and rotation, displacement-coupled mobilized earth-pressure updating, elastic base restraint, staged nonlinear reinforcement/support interaction, and energy-compatible equilibrium iteration for complex soil–structure interaction problems.

This is currently the most advanced solver.

It is a real nonlinear beam finite-element formulation governed by:

$$[K_{beam} + K_{base} + K_{support}]w = P_{soil}(w) + P_{support}$$

Main characteristics:

- Euler–Bernoulli beam stiffness
- nodal rotational DOFs
- nonlinear CUT pressures
- elastic base springs
- unilateral supports
- secant stiffness caps
- staged support activation
- true beam compatibility

Supports are represented as actual stiffness elements.

Iterative nonlinear solution:

1. Compute mobilized pressures from current wall deflection
2. Assemble nonlinear beam system
3. Solve FE equations
4. Update supports
5. Apply adaptive relaxation
6. Repeat until convergence

Optimization/stabilization:

- nonlinear fixed-point iteration
- adaptive relaxation
- oscillation detection
- secant support capping
- energy diagnostics

Energy compatibility:

$$W_{ext} \approx U_{beam} + U_{support}$$

This is currently the flagship solver of CUT Embedded Wall and the closest formulation within the platform to a true nonlinear soil–structure interaction analysis framework.

4. Support and Reinforcement Systems

- Anchored walls
- Propped, braced, and strutted wall systems
- Multi-level support configurations
- True nodal reinforcement/support formulation
- Nonlinear one-way support interaction
- Modeling of support stiffness, capacity, spacing, free length, inclination, and prestress

5. Visualization and Engineering Workflow

- Real-scale geometry and reinforcement visualization
- High-quality engineering plots and animations

6. Core Concept

CUT Embedded Wall is built around the idea that many retaining wall problems can be treated with a high level of physical rigor **without abandoning analytical transparency**.

The software does not attempt to replace full finite element analysis in every possible situation. Instead, it provides a specialized analytical framework for earth pressure and embedded wall problems, where the governing mechanics are exposed, traceable, and computationally efficient.

This is particularly important because many conventional design-code approaches to earth pressures do not use a unified theoretical framework. They often rely on separate formulations for different states, intuitive correction factors, semi-empirical additions, and in some cases mathematical complications such as negative roots or nonphysical solution branches.

CUT Embedded Wall addresses these issues by using a unified earth pressure engine rather than a patchwork of disconnected expressions.

7. The Engine: Pantelidis Generalized Coefficients of Earth Pressure

At the heart of CUT Embedded Wall is the Pantelidis Generalized Coefficients of Earth Pressure framework.

This framework introduces a unified continuous formulation for describing the progressive mobilization of earth pressures, from at-rest conditions through intermediate mobilized states toward active or passive failure.

The framework allows continuous transition between:

- at-rest conditions,
- active mobilization,
- passive mobilization,
- and intermediate mobilized states.

It simultaneously incorporates:

- cohesion,
- friction,
- pseudo-static seismic loading,
- surcharge effects,
- and nonlinear mobilization behavior.

8. Why the K_{XE} Engine Is Important

The generalized coefficient engine is one of the central innovations of the program.

It offers a unified framework that:

- connects K_0 , K_a , and K_p ,
- includes cohesion,
- incorporates pseudo-static seismic loading,
- handles cohesive-frictional soils,
- avoids artificial switching between unrelated formulas,
- supports displacement-dependent mobilization,
- and provides a consistent basis for both retaining wall and pile problems.

This is a major departure from conventional earth pressure approaches, where active, passive, at-rest, cohesive, and seismic cases are frequently handled by separate equations or additional correction terms.

The engine, therefore, provides a coherent mechanical basis for a wide range of geotechnical problems involving earth pressures and soil reactions.

9. Validation of the Engine

The Pantelidis generalized earth pressure methodology has been extensively validated.

It has been checked against:

- hundreds of finite element models,
- independent numerical comparisons,
- centrifuge test data,
- seismic earth pressure benchmarks,
- retaining wall cases,
- axially loaded pile problems,
- laterally loaded pile problems.

This broad validation background is one of the main reasons the engine can be used as the central component of CUT Embedded Wall.

The same generalized coefficient concept has been extended beyond retaining walls to problems such as:

- embedded wall design,
- shaft resistance of axially loaded piles,
- laterally loaded piles,
- static and pseudo-static soil–structure interaction,
- and generalized lateral soil reaction problems.

10. Supported Analysis Configurations

CUT Embedded Wall currently supports the analysis of:

- cantilever embedded retaining walls,
- anchored retaining wall systems,
- propped and braced excavation systems,
- and multi-level reinforced wall configurations.

The framework is based on:

- flexible wall behavior,
- beam-type and differential-equation structural formulations,

- and displacement-coupled soil–structure interaction.

The wall response is calculated through:

- bending-based structural analysis,
- iterative mobilization of earth pressures,
- and nonlinear reinforcement/support interaction.

The current implementation includes:

- fixed-base flexible wall analysis,
- nonlinear anchor interaction,
- nonlinear prop and strut interaction,
- support stiffness and capacity effects,
- prestress support modeling,
- support spacing, inclination, and geometric effects,
- and generalized reinforcement configurations within a unified computational framework.

The modular architecture also allows future extension toward additional analytical formulations, advanced soil–structure interaction models, and expanded reinforcement and boundary-condition capabilities.

11.Nonlinear Soil–Structure Interaction

CUT Embedded Wall couples:

- wall deformation,
- mobilized earth pressures,
- structural bending,
- and support forces.

The pressure field is not simply a fixed active or passive diagram. Instead, mobilized pressures are recalculated from the current displacement profile of the wall.

This allows the program to represent:

- partial mobilization,
- pressure redistribution,
- gradual support engagement,
- reduction of bending due to reinforcement,
- and the development of realistic deformation mechanisms.

12.Reinforcement and Support Modeling

Supports are not treated merely as pressure corrections.

Anchors, props, and struts are modeled as support elements interacting with the wall deformation.

The support model can include:

- axial stiffness,
- unsupported or free length,
- spacing,
- inclination,
- prestress,
- capacity,
- tension-only behavior,
- compression-only behavior,
- utilization checks,
- nodal reaction recovery.

This makes it possible to compare unsupported, anchored, and propped systems in a consistent way.

13.Engineering Visualization

The software produces advanced engineering plots and visualization outputs including:

- total earth pressure distributions,
- net pressure distributions,
- wall deflection profiles,
- wall rotation profiles,
- shear force diagrams,
- bending moment diagrams,
- K —diagrams,
- normalized displacement and mobilization diagrams,
- convergence histories,
- reinforcement and support force distributions and tables

The platform also supports true-scale geometry and reinforcement visualization, allowing wall depth, excavation geometry, support locations, and reinforcement lengths to be interpreted directly at true physical scale.

Visualization outputs are dynamically coupled to the analytical–computational engine, enabling realistic interpretation of soil–structure interaction, pressure redistribution, support engagement, and deformation mechanisms.

14. Advanced Visualization and Animation

CUT Embedded Wall includes advanced engineering visualization and animation capabilities designed to improve interpretation of soil–structure interaction mechanisms and staged excavation behavior.

The platform supports:

- animated wall deformation development,
- staged excavation and support activation visualization,
- progressive mobilization of earth pressures,
- dynamic visualization of active, passive, and at-rest pressure evolution,
- water-level variation animations,
- reinforcement/support force evolution,
- convergence history visualization,
- true-scale geometry rendering,
- synchronized deformation–pressure–moment interaction plots.

The animation framework is directly coupled to the analytical–computational engine rather than post-processed independently. This allows the visualization to reflect the actual iterative mobilization process and nonlinear soil–structure interaction response of the system.

The visualization environment is intended not only for presentation purposes, but also as an engineering interpretation tool for understanding:

- deformation mechanisms,
- support engagement,
- pressure redistribution,
- stiffness effects,
- and nonlinear interaction behavior.

15. Streamlit Interface

The Streamlit version provides:

- clean and intuitive model input forms,
- editable engineering tables,
- interactive results dashboards,
- real-time engineering plots and visualization,
- automatic PDF report generation,
- CSV export capabilities,
- integrated manual/documentation access,
- modern browser-based engineering workflows,
- and cross-platform usability across Windows, macOS, Linux, iOS, and Android devices.

16.Intended Use

CUT Embedded Wall is intended for:

- teaching,
- academic research,
- method development,
- independent checking,
- parametric studies,
- comparison with finite element models,
- development of new geotechnical formulations.

It is especially useful where the engineer or researcher wants to understand the mechanics rather than simply obtain a black-box answer.

17.Related Papers

- Pantelidis, L. (2019). The generalized coefficients of earth pressure: a unified approach. *Applied Sciences*, 9(24), 5291.
- Pantelidis, L. From EN 1998-5: 2004 to prEN 1998-5: 2023: Has the calculation of earth pressures improved or deteriorated?
- Pantelidis, L. (2023, November). An Analytical Method for Designing Embedded Retaining Walls. In *Rocscience International Conference (RIC 2023)*, pp. 217–224. Atlantis Press.
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- Pantelidis, L. (2023, November). An Analytical Method for Designing Laterally Loaded Piles. In *Rocscience International Conference (RIC 2023)*, pp. 567–574. Atlantis Press.
- Pantelidis, L. (2023). Designing laterally loaded piles relying on the Generalized Coefficient of Earth Pressure and static analysis of elastic beams. *Research Square*.
- Pantelidis, L., & Christodoulou, P. (2022). Comparing Eurocode 8-5 and AASHTO methods for earth pressure analysis against centrifuge tests, finite elements, and the Generalized Coefficients of Earth Pressure. *Research Square*, 3, 1–52.
- Pantelidis, L. (2022). Shaft resistance capacity of axially loaded piles in cohesive-frictional soils under static or pseudo-static conditions based on ground parameters.
- Pantelidis, L., Christodoulou, P., Assefa, E., & Sachpazis, C. (2021, November). The generalized coefficients of earth pressure: Numerical validation and comparison with Eurocode 8-5 method. In *International Conference on Mediterranean Geosciences Union*, pp. 47–51. Springer Nature Switzerland.

18.Disclaimer

CUT Embedded Wall is educational and research software.

It is not a certified commercial design package. No warranty is provided regarding correctness, completeness, fitness for design, numerical stability under all conditions, or compliance with any specific design code.

All results must be reviewed and independently verified by qualified engineers before practical use.